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#include "MKL25Z4.h"

#define ADC_MAX_VALUE 4095U #define TH_VALUE (ADC_MAX_VALUE * 80U / 100U) #define TL_VALUE (ADC_MAX_VALUE * 20U / 100U)

#define LED_BLUE_PIN 1 #define LED_GREEN_PIN 19 #define ADC_CHANNEL 8

void config_leds(void); void config_adc(void); uint16_t read_adc(uint8_t channel);

int main(void) { config_leds(); config_adc();

```
while(1)
{
    uint16_t value_adc = read_adc(ADC_CHANNEL);

    if (value_adc >= TH_VALUE)
    {
        GPIOB->PSOR = (1 << LED_GREEN_PIN);
        GPIOD->PCOR = (1 << LED_BLUE_PIN);
    }
    else if (value_adc <= TL_VALUE)
    {
        GPIOB->PCOR = (1 << LED_GREEN_PIN);
        GPIOD->PSOR = (1 << LED_BLUE_PIN);
    }
    else
    {
        GPIOB->PSOR = (1 << LED_GREEN_PIN);
        GPIOD->PSOR = (1 << LED_BLUE_PIN);
    }
}

return 0;
```

}

void config_leds(void) { SIM->SCGC5 |= SIM_SCGC5_PORTD_MASK; PORTD->PCR[LED_BLUE_PIN] = PORT_PCR_MUX(1);
GPIOD->PDDR |= 1 << LED_BLUE_PIN;
GPIOD->PSOR = 1 << LED_BLUE_PIN;

```
SIM->SCGC5 |= SIM_SCGC5_PORTB_MASK;
PORTB->PCR[LED_GREEN_PIN] = PORT_PCR_MUX(1);
GPIOB->PDDR |= 1 << LED_GREEN_PIN;
GPIOB->PSOR = 1 << LED_GREEN_PIN;
```

}

void config_adc(void) { SIM->SCGC5 |= SIM_SCGC5_PORTB_MASK; PORTB->PCR[0] = PORT_PCR_MUX(0);
SIM->SCGC6 |= SIM_SCGC6_ADC0_MASK;

```
ADC0->CFG1 = ADC_CFG1_ADIV(1) | ADC_CFG1_MODE(1) | ADC_CFG1_ADICLK(0);
ADC0->CFG2 = ADC_CFG2_ADLSTS(3);
ADC0->SC2 = 0;
ADC0->SC3 = 0;
```

}

uint16_t read_adc(uint8_t channel) { ADC0->SC1[0] = ADC_SC1_ADCH(channel); while(!(ADC0->SC1[0] & ADC_SC1_COCO_MASK)); return ADC0->R[0]; }

https://github.com/Arthur270903/Embarcados-